THE TOR GROUPS OF THE INJECTIVE HULL AGAINST THE RESIDUE FIELD OVER A GORENSTEIN RING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of the local cohomology of the normalized dualizing complex as the injective hull in degree zero. Invoked inside the counted proof (source lines 4231-4234). Identifies the injective hull with that local cohomology.
- The interaction of local cohomology with derived tensor products. Invoked inside the counted proof (source lines 4235-4238). Allows the tensor with the residue field to be moved inside.

Source reference. Original lemma lemma-local-cohomology-of-dualizing: sources/stacks-dualizing/pinned-original-source.tex, source lines 3642–3642. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-of-dualizing, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-torsion-tensor-product: sources/stacks-dualizing/pinned-original-source.tex, source lines 1627–1627. Navigation pointer only: the counted text refers to the Stacks item lemma-torsion-tensor-product, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian local Gorenstein ring of dimension d. Let E be the injective hull of κ. Then $Tor_i^A(E, \kappa)$ is zero for $i \neq d$ and $Tor_d^A(E, \kappa) = \kappa$.*

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Proof. Since A is Gorenstein $\omega_A^\bullet = A[d]$ is a normalized dualizing complex for A. Also E is the only nonzero cohomology module of $R\Gamma_{\mathfrak{m}}(\omega_A^\bullet)$ sitting in degree 0, see Lemma lemma-local-cohomology-of-dualizing. By Lemma lemma-torsion-tensor-product we have

$$E \otimes_A^{\mathbf{L}} \kappa = R\Gamma_{\mathfrak{m}}(\omega_A^\bullet) \otimes_A^{\mathbf{L}} \kappa = R\Gamma_{\mathfrak{m}}(\omega_A^\bullet \otimes_A^{\mathbf{L}} \kappa) = R\Gamma_{\mathfrak{m}}(\kappa[d]) = \kappa[d]$$

and the lemma follows. □